

Software Testing Documentation

StockLite Mobile (STOCKLITE-MOBILE) – Report 5

Version	1.0
Date	21/07/2026

Change Log

Version	Date	Author	Summary
1.0	21/07/2026	QA / Development Team	Replaced template examples with StockLite Mobile test scope, strategy, plan, cases, and execution results.

Table of Contents

Change Log.....	1
Table of Contents	2
1. Scope of Testing	3
1.1 In Scope - Functional Requirements.....	3
1.2 In Scope - Non-Functional Requirements.....	4
1.3 Out of Scope.....	4
1.4 Constraints	5
1.5 Assumptions	5
2. Test Strategy	6
2.1 Testing Types	6
2.2 Test Levels	6
a. Automated Test Flow.....	7
b. Manual BLE Flow	7
2.3 Supporting Tools	7
3. Test Plan	9
3.1 Test Environment	9
3.2 Test Data.....	9
a. Minimum Test Accounts.....	9
b. Minimum Domain Data.....	9
3.3 Test Milestones	10
4. Test Cases	11
4.1 Automated Test Assets	11
4.2 Requirements Coverage Matrix.....	12
5. Test Reports	13
5.1 Execution Summary	13
5.2 Defect Register	13
5.3 Exit Criteria and Recommendation.....	13

1. Scope of Testing

This test cycle validates the StockLite Mobile application used by field staff for purchasing, warehouse operations, delivery, stocktake, quality inspection, milling, reporting, QR scanning, and account management. The scope covers Flutter business logic, screen behavior, API integration, Android execution, and the BLE scale workflow.

1.1 In Scope - Functional Requirements

Code	Feature / Requirement	Priority
FT-01	Authenticate with the REST API, retain the authenticated session, display API errors, and reject invalid credentials.	Must
FT-02	Display Home dashboard, today summary, quick actions, and bottom navigation.	Must
FT-03	Display the authenticated account profile, change password, operation history, and sign out.	Must
FT-04	Load purchase schedules, open schedule details, and execute the inbound purchasing flow.	Must
FT-05	Display only products with on-hand stock and open warehouse product details.	Must
FT-06	Create stocktake entries and accept multi-digit actual quantities.	Must
FT-07	Keep quality inspection separate from stocktake and display the correct workflow.	Must
FT-08	Load delivery data, open delivery details, select customer/stock, and complete delivery.	Must
FT-09	Run milling preparation, rice weighing, bran weighing, result confirmation, and completion.	Must
FT-10	Scan, connect, read stable weight, tare, reset, and capture multiple bags from the StockLite BLE scale.	Should
FT-11	Display warehouse reports and operation history from repository/API data.	Should
FT-12	Open the QR/barcode scanner and handle camera permission safely.	Should
FT-13	Load REST notifications and refresh them manually without realtime dependencies.	Could

1.2 In Scope - Non-Functional Requirements

Code	Category	Description / Target
NFR-C01	Compatibility	Build and run on Android 15 / API 35 emulator; retain compatibility with supported physical Android devices.
NFR-R01	Reliability	No crash or blank screen while navigating the automated smoke-test scope.
NFR-U01	Usability	Loading, empty, success, and error states are understandable and do not require manual state selection.
NFR-S01	Security	Credentials are not stored in test source; protected requests use the access token returned by the login API.
NFR-B01	Bluetooth	Only stable BLE readings with weight greater than zero can create milling bag records.
NFR-T01	Testability	All committed unit/widget tests and mobile integration smoke tests must pass before delivery.

1.3 Out of Scope

Item	Reason
Firebase Cloud Messaging and SignalR realtime notifications	Removed from the current mobile release and scheduled for a later implementation.
Formal load, stress, and penetration testing	Requires dedicated infrastructure and a controlled backend test environment.
iOS release and App Store deployment	Current verification target is Android.
Automated physical BLE hardware validation	An emulator cannot reliably scan the ESP32 scale; final validation is manual on a real Android phone.
Backend internal unit tests	The backend repository is not modified by this mobile test cycle.

1.4 Constraints

Category	Constraint
Backend	The default API is hosted on Render and may have cold-start latency.
BLE hardware	StockLite scale tests require ESP32/HX711 hardware and a physical Android device with Bluetooth permissions.
Emulator	Android emulator supports UI/API tests but normally cannot discover the BLE scale.
Milling API	The backend does not expose a dedicated endpoint for each captured bag; individual readings remain local until completion.
Test data	Availability and state of schedules, stock, customers, deliveries, and milling orders depend on backend seed data.

1.5 Assumptions

#	Assumption
1	The external API is reachable and the supplied test account remains active.
2	Test data can be changed without affecting production business records.
3	Android camera and nearby-device permissions can be granted or denied during testing.
4	The ESP32 firmware advertises the documented StockLite service and characteristic UUIDs.
5	A failed test is rerun after confirming network, backend availability, and seed-data state.

2. Test Strategy

Testing follows a risk-based approach. Must-priority field operations and authentication receive automated regression coverage first. Hardware-specific and data-dependent scenarios are completed through structured manual tests on a physical Android device.

2.1 Testing Types

Type	Coverage	Execution
Unit testing	JSON conversion, milling totals/yield, copyWith behavior, and BLE payload parsing. Automated with flutter_test	
Widget testing	Screen rendering, validation messages, account data, navigation, stocktake input, and workflow behavior.	Automated with flutter_test
Mobile integration testing	Application startup, login form interaction, and login against the configured external API.	Automated on Android emulator/device
API functional testing	Repository/API flows for authentication and mobile business screens.	Automated where deterministic; otherwise manual
BLE testing	Permissions, scanning, connection, notify payload, stable capture, tare/reset, reconnect, and multiple bags.	Manual on physical Android device
Regression testing	Full Flutter test suite and smoke integration tests after meaningful code changes.	Automated before handoff

2.2 Test Levels

Level	Scope	Current Result	Exit Criteria
L1 - Unit	Pure model/helper logic without device or network.	15 focused logic tests pass; included in 32-test Flutter suite.	All tests pass.
L2 - Widget	Screens, inputs, states, and navigation with fake repositories/services.	All widget tests pass.	No failed widget regression.
L3 - Mobile Integration	Compiled application on Android emulator/device.	Mobile smoke and external API login tests pass.	App launches and critical login path succeeds.
L4 - System / Manual	End-to-end business flows with live backend and hardware.	Partially executed; BLE requires physical device.	Critical flows pass with representative data.

Level	Scope	Current Result	Exit Criteria
L5 - UATTester/business acceptance of field workflows. Critical/Major blocker and signed acceptance.		Pending formal sign-off.	No open

a. Automated Test Flow

- Run flutter test for all unit and widget tests.
- Run integration_test/mobile_smoke_test.dart on emulator or physical device.
- Run integration_test/api_login_test.dart with credentials supplied through dart-define.
- Build the Android debug APK and install it with ADB.
- Review failures with Flutter output, screenshots, and Android logcat.

b. Manual BLE Flow

- Use a physical Android phone; enable Bluetooth and grant Nearby Devices permission.
- Confirm ESP32 advertises StockLite SCALE-01 and the documented service UUID.
- Capture stable rice and bran readings and verify bag count and total weight.
- Exercise tare, reset, disconnect, reconnect, permission denial, and Bluetooth-off behavior.

2.3 Supporting Tools

Purpose	Tool	Version / Configuration
Application framework	Flutter	3.41.9 stable
Programming language	Dart	3.11.5
Unit/widget testing	flutter_test	Flutter SDK
Mobile integration testing	integration_test	Flutter SDK
Android device control	ADB / Android SDK	SDK 36.1; emulator API 35
BLE communication	flutter_blue_plus	2.3.10
Runtime permissions	permission_handler	12.0.3
Static analysis	flutter analyze	Flutter SDK

Purpose	Tool	Version / Configuration
Test automation scripts	PowerShell	prepare_mobile_test.ps1 / run_mobile_tests.ps1

3. Test Plan

3.1 Test Environment

Area	Configuration	Purpose
Development host documentation.	Windows 11 Enterprise 64-bit	Build, unit/widget tests, ADB, and
Flutter runtime	Flutter 3.41.9 / Dart 3.11.5	Compile and execute StockLite Mobile.
Android emulator	Android 15, API 35, emulator-5554 Automated UI/API mobile integration tests.	
Physical Android phone	Bluetooth Low Energy capable device	BLE scale and permission testing.
External API	https://backend-do-an-api-new.onrender.com	Default integrated backend.
Local API option	10.0.2.2:PORT for emulator; PC LAN IP for phone	Developer/local backend verification.
BLE device	ESP32 + HX711, StockLite SCALE-01	Realtime milling weight readings.

3.2 Test Data

a. Minimum Test Accounts

Username	Role	Password Handling	Purpose
admin	Administrator	Provided securely at execution time	Login API, account information, and privileged flows.
nhanvienkho	Warehouse staff	Provided securely at execution time	Warehouse, stocktake, delivery, and field workflows.
invalid-user	None	Deliberately invalid	Negative authentication and error-message tests.

b. Minimum Domain Data

Entity	Required State	Purpose
--------	----------------	---------

Entity	Required State	Purpose
Purchase schedule flow.	At least one accessible schedule with detail data	Purchase detail and inbound
Warehouse stock	At least one product with quantity greater than zero	Warehouse list, product detail, delivery, and stocktake.
Customer	At least one active customer	Delivery creation and detail validation.
Delivery	At least one completed or viewable delivery	Delivery detail regression.
Milling order completion flow.	At least one active order with input lot/location	Preparation, BLE weighing, and
ESP32 scale	Advertising and sending valid JSON weight payloads	Physical BLE validation.

3.3 Test Milestones

Milestone	Status	Owner	Acceptance Criteria
M1 - Unit/widget regression	Completed	Development / QA	32 of 32 Flutter tests pass.
M2 - Mobile smoke integration	Completed	QA	Application opens and login fields work on Android emulator.
M3 - External API login integration	Completed	QA	Configured account reaches Home through the hosted backend.
M4 - Full manual business flow	Ready	Mobile tester	Critical purchasing, warehouse, stocktake, delivery, and milling flows pass.
M5 - Physical BLE acceptance	Pending hardware execution	Mobile tester	Scale discovery, stable capture, commands, and reconnect pass.
M6 - UAT sign-off	Pending Product owner / tester	No Critical/Major blocker and written acceptance.	

4. Test Cases

4.1 Automated Test Assets

Test File	Cases	Primary Coverage
account_tab_test.dart	1	Authenticated user name, email, avatar initial, and removal of hard-coded profile.
all_screens_smoke_test.dart	4	Reports, inbound/outbound success, and QR scanner shell.
giao_hang_detail_test.dart	1	Completed delivery detail.
inspection_separation_test.dart	1	Quality inspection and stocktake route separation.
json_reader_test.dart	4	Case-insensitive keys and safe type conversion.
kho_inventory_test.dart	2	Positive on-hand filtering and latest stocktake date.
login_screen_error_test.dart	2	Incorrect username/password handling without creating a session.
milling_flow_test.dart	3	Milling calculations and navigation through weighing/confirmation.
milling_order_test.dart	4	Rice/bran totals, yield, zero input, and copyWith.
purchase_schedule_detail_test.dart	1	Purchase schedule detail mobile layout.
stocktake_multi_digit_input_test.dart	1	Multi-digit actual quantity input.
weight_reading_test.dart	7	Valid and invalid ESP32 BLE JSON payloads.
widget_test.dart	1	Login screen branding and basic content.
integration_test/mobile_smoke_test.dart	1	Compiled mobile startup and login input.
integration_test/api_login_test.dart	1	Hosted API login and navigation to Home.

4.2 Requirements Coverage Matrix

FT-ID	FeatureUnit / Widget	Integration	Manual
FT-01	Authentication	Covered	Covered Required
FT-02	Home/dashboard	Covered	Smoke Required
FT-03	Account/session	Covered Partial	Required
FT-04	Purchasing/inbound	Covered	Data-dependent Required
FT-05	Warehouse/product detail	Covered	Data-dependent Required
FT-06	Stocktake	Covered	Data-dependent Required
FT-07	Quality inspection	Covered	Data-dependent Required
FT-08	Delivery	Covered	Data-dependent Required
FT-09	Milling workflow	Covered	Data-dependent Required
FT-10	BLE scale	Parser covered	Not on emulator Physical device required
FT-11	Reports/history	Partial	Data-dependent Required
FT-12	QR scanning	Shell covered	Permission-dependent Physical camera required
FT-13	REST notifications	Partial	Data-dependent Required

5. Test Reports

5.1 Execution Summary

Round	Level	Executed	Passed	Failed	Status
Round 1	Unit + Widget	32	32	0	Passed
Round 1	Mobile Smoke Integration	1	1	0	Passed
Round 1	External API Login Integration	1	1	0	Passed
Round 1	Physical BLE / Manual	0	0	0	Pending
Total executed	Automated	34	34	0	Passed

5.2 Defect Register

ID	Observation / Defect	Severity	Status	Action
OBS-01	Flutter analyzer reports information-level style/deprecation notices in existing code. Open observation	Clean up incrementally; no compile or test failure.	Low	
OBS-02	Hosted Render backend may respond slowly after an idle cold start. Allow longer timeout and distinguish cold start from functional failure.	Low	Known limitation	
OBS-03	BLE discovery cannot be accepted on the Android emulator. hardware test Execute the BLE checklist on a physical Android phone.	Expected constraint	Pending	
OBS-04	Backend has no endpoint dedicated to persisting every individual milling bag reading. Backend gap Keep readings local and define an API contract before production persistence.	Medium		

5.3 Exit Criteria and Recommendation

- Automated status: PASS - 34 of 34 executed automated tests passed.
- Android build and emulator execution are operational.
- Proceed to structured manual testing of live backend business data.
- Do not mark BLE scale acceptance complete until it passes on a physical Android phone and ESP32 scale.
- Release recommendation: conditionally ready for tester validation; final acceptance depends on manual business-flow and BLE evidence.

Prepared from the StockLite Mobile codebase and automated test results available on 21/07/2026.